

HQ-to-Branch Network and Active Directory Integration

Objective

Configure and validate routed communication between the **BENLAB Headquarters network** and the simulated **Branch 01 network**, allowing branch systems to communicate securely with headquarters infrastructure and Active Directory services.

The implementation required configuring routing between two pfSense firewalls, validating bidirectional connectivity, troubleshooting firewall state behavior, permitting required Active Directory services, configuring branch DNS appropriately, and validating domain communication from multiple branch systems.

The final objective was to establish reliable communication between:

Headquarters

- Network: 10.20.20.0/24
- Firewall: pfSense01
- Domain Controller/DNS: DC01 - 10.20.20.10
- Domain: benlab.local

Branch 01

- Network: 10.30.30.0/24
- Firewall: BRANCHFW01 - 10.30.30.254
- Client: CLIENT02 - 10.30.30.100
- Server: FILESRV02 - 10.30.30.101

Environment and Network Topology

The lab simulates two physically separate business locations connected through a routed transit network.

BENLAB HEADQUARTERS
10.20.20.0/24

DC01
10.20.20.10

```

|
|
pfSense01
LAN: 10.20.20.254
WAN: 192.168.0.29
|
|
192.168.0.0/24
Transit Network
|
|
BRANCHFW01
WAN: 192.168.0.28
LAN: 10.30.30.254
|
|
BRANCH 01
10.30.30.0/24
|
+---- CLIENT02
|      10.30.30.100
|
+---- FILESRV02
|      10.30.30.101

```

VMware Host-only networks provide the isolated HQ and Branch LANs, while bridged pfSense WAN interfaces provide connectivity across the physical transit network.

Static Routing Configuration

Static routes were configured so each pfSense firewall knew how to reach the remote internal network.

HQ – pfSense01

A route was configured for:

```

Destination Network: 10.30.30.0/24
Gateway: BRANCHFW01
Gateway Address: 192.168.0.28

```

Branch – BRANCHFW01

The Branch firewall was configured to reach:

```

Destination Network: 10.20.20.0/24
Gateway: pfSense01
Gateway Address: 192.168.0.29

```

Route-table testing verified that traffic destined for the remote LAN was being directed through the appropriate pfSense WAN interface.

Initial Connectivity Troubleshooting

Initial testing revealed **asymmetric connectivity** between the two locations.

Some traffic could travel from HQ toward Branch, but return traffic did not consistently reach HQ.

Testing included:

```
ping
route
arp
tcpdump
pktmon
Test-NetConnection
```

Rather than assuming that a successful ping proved the network was fully operational, connectivity was validated progressively at multiple layers.

pfSense WAN Private-Network Configuration

The pfSense WAN interfaces use private RFC1918 addresses because the WAN network is a simulated internal transit network:

```
192.168.0.0/24
```

During troubleshooting, the pfSense WAN option:

```
Block private networks and loopback addresses
```

was identified as interfering with communication arriving from the private transit network.

Because this WAN is intentionally being used for an internal lab connection, the private-network blocking option was disabled where appropriate.

After the change, previously failing WAN-to-WAN communication succeeded.

VMware Bridged Networking Validation

Both pfSense WAN interfaces depend on VMware bridged networking to communicate across the physical network.

VMware VMnet0 was explicitly associated with the active physical:

```
Realtek PCIe GbE Family Controller
```

rather than relying on automatic bridging.

This removed ambiguity about which physical adapter VMware was using for the pfSense WAN transit network.

Packet-Level Troubleshooting

Packet captures were used extensively to determine exactly where traffic was being lost.

On pfSense, tcpdump and the graphical **Diagnostics** → **Packet Capture** utility were used.

On Windows, Packet Monitor (pktmon) was used.

Example:

```
pktmon start --capture --comp nics --pkt-size 0
```

After generating the test traffic:

```
pktmon stop  
pktmon etl2txt PktMon.etl
```

The resulting text capture was filtered for specific IP addresses.

Example:

```
Select-String -Path .\PktMon.txt -Pattern "10.30.30.100"
```

Packet captures eventually demonstrated that CLIENT02 was receiving packets from HQ and generating replies.

This helped eliminate the Windows endpoint as the sole cause and moved troubleshooting toward the firewall/routing path.

pfSense Stateful Firewall and reply-to

One of the most significant findings involved pfSense's stateful firewall behavior.

Firewall states showed traffic entering the routed path, but return traffic was not always following the expected static route.

Advanced firewall-rule configuration was investigated, particularly:

```
Disable reply-to
```

The inter-site rules were configured with:

```
Disable auto generated reply-to for this rule
```

enabled.

This was particularly important because the pfSense WAN interfaces have their own default gateway while the BENLAB inter-site networks use dedicated static routes.

Disabling automatically generated `reply-to` on the applicable inter-site rules allowed the normal routing table/static route to determine the correct return path.

After correcting the rule behavior and resetting firewall states, previously failing inter-site TCP communication began succeeding.

Active Directory Service Validation

Successful ICMP communication alone was not considered sufficient proof of Active Directory connectivity.

Individual services required by Active Directory were therefore tested from CLIENT02 toward:
DC01 - 10.20.20.10

PowerShell testing used:

```
Test-NetConnection 10.20.20.10 -Port <port>
```

Examples included LDAP, SMB, RPC Endpoint Mapper and dynamically allocated RPC services.

Important services encountered during troubleshooting included:

TCP 389	LDAP
TCP 445	SMB / Microsoft-DS
TCP 135	RPC Endpoint Mapper
TCP 49152-65535	Dynamic RPC range

Additional AD-related services were validated as required during the troubleshooting process.

A particularly important observation was that a system could show:

```
PingSucceeded      : True  
TcpTestSucceeded  : False
```

This demonstrated that **Layer 3 reachability did not guarantee application/service connectivity**.

RPC Dynamic Port Troubleshooting

RPC Endpoint Mapper connectivity alone was not sufficient for all Active Directory operations.

DC01 was examined for dynamically allocated listening ports:

```
Get-NetTCPConnection -State Listen |  
Where-Object { $_.LocalPort -ge 49152 } |  
Sort-Object LocalPort |  
Select-Object -First 10 LocalAddress, LocalPort, State
```

This identified active RPC dynamic ports, including ports in the Windows dynamic range.

Branch-to-HQ firewall access was configured for:

```
TCP 49152-65535
```

with the appropriate inter-site source and destination restrictions.

After the rule was applied and `Disable reply-to` was correctly configured, dynamic RPC connectivity succeeded.

Branch DNS Troubleshooting

CLIENT02 originally used the Branch firewall as its DNS server:

```
10.30.30.254
```

Although normal network communication was possible, Active Directory DC Locator queries failed.

An SRV query demonstrated the issue:

```
nslookup -type=SRV _ldap._tcp.dc._msdcs.benlab.local
```

The Branch DNS service returned:

```
Non-existent domain
```

CLIENT02 was then configured to use the BENLAB DNS server hosted on DC01:

```
Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses 10.20.20.10
```

The configuration was verified with:

```
Get-DnsClientServerAddress -InterfaceAlias "Ethernet0" -AddressFamily IPv4
```

Result:

```
ServerAddresses
-----
{10.20.20.10}
```

This allowed CLIENT02 to query the authoritative BENLAB DNS infrastructure directly.

Domain Controller Discovery

Domain Controller discovery was tested using:

```
nlttest /dsgetdc:benlab.local
```

Before DNS and service connectivity were corrected, the command returned:

```
ERROR_NO_SUCH_DOMAIN
```

After correcting the configuration, CLIENT02 successfully discovered:

```
DC: \\DC01.benlab.local
Address: \\10.20.20.10
Domain Name: benlab.local
Forest Name: benlab.local
```

The command completed successfully and identified DC01 capabilities including:

```
PDC
GC
DS
LDAP
KDC
TIMESERV
WRITABLE
DNS_DC
DNS_DOMAIN
DNS_FOREST
```

This provided significantly stronger validation than a simple ping test because Windows was successfully using **Active Directory DC Locator**.

CLIENT02 Secure Channel Repair

CLIENT02 had previously been joined to benlab.local, but its machine trust relationship was no longer healthy.

Testing showed:

```
Test-ComputerSecureChannel -Verbose
```

Result:

```
False
The secure channel between the local computer and the domain benlab.local is broken.
```

An initial repair attempt failed because all required domain services were not yet available across the routed connection.

After DNS, firewall, RPC and routing issues were resolved, the secure channel was repaired using:

```
Test-ComputerSecureChannel -Repair -Credential (Get-Credential) -Verbose
```

The repair completed successfully.

Final validation:

```
Test-ComputerSecureChannel -Verbose
```

Result:

True

The secure channel between the local computer and the domain benlab.local is in good condition.

FILESRV02 Branch Server Integration

After CLIENT02 was successfully validated, FILESRV02 was used as a second Branch endpoint to ensure that the solution was not limited to a single workstation.

FILESRV02 received:

IPv4 Address: 10.30.30.101
Subnet Mask: 255.255.255.0
Default Gateway: 10.30.30.254
DHCP Server: 10.30.30.254

Its DNS configuration was changed to:

10.20.20.10

DC Locator testing was then performed:

nltest /dsgetdc:benlab.local

FILESRV02 successfully discovered:

DC: \\DC01.benlab.local
Address: \\10.20.20.10

This demonstrated that the Branch server could discover Active Directory services across the routed connection.

FILESRV02 Domain Join

FILESRV02 was not previously a member of the BENLAB domain.

It was joined using PowerShell:

Add-Computer -DomainName benlab.local -Credential (Get-Credential) -Restart

After restart, domain trust was validated with:

Test-ComputerSecureChannel -Verbose

Result:

True

The secure channel between the local computer and the domain benlab.local is in good condition.

DC discovery was tested again:

```
nltest /dsgetdc:benlab.local
```

Result:

DC: \\DC01.benlab.local

Address: \\10.20.20.10

...

The command completed successfully.

This provided a second independent validation that Branch systems could successfully consume Headquarters Active Directory services.

Final Validation

The following functionality was successfully demonstrated:

HQ ↔ Branch routing

Traffic successfully traverses:

10.20.20.0/24 ↔ 10.30.30.0/24

through:

pfSense01 ↔ Transit Network ↔ BRANCHFW01

Active Directory service connectivity

Branch systems can reach required services hosted by DC01 across the routed network.

DNS

Branch domain systems can use:

DC01 – 10.20.20.10

for BENLAB Active Directory DNS resolution.

Domain Controller discovery

Both CLIENT02 and FILESRV02 successfully discover:

DC01.benlab.local

using Windows DC Locator.

CLIENT02

Secure channel:

True

FILESRV02

Successfully joined:

benlab.local

Secure channel:

True

Key Troubleshooting Commands

`ipconfig /all`

Displays detailed IP, gateway, DHCP, DNS and adapter configuration.

`Test-NetConnection 10.20.20.10 -Port 389`

Tests TCP connectivity to an individual remote service.

`Get-DnsClientServerAddress -AddressFamily IPv4`

Displays the DNS servers configured on Windows interfaces.

`Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses 10.20.20.10`

Changes the DNS server used by the selected Windows interface.

`nslookup -type=SRV _ldap._tcp.dc._msdcs.benlab.local`

Tests the Active Directory DNS SRV records used for Domain Controller discovery.

`nltest /dsgetdc:benlab.local`

Forces Windows to locate a Domain Controller for the specified domain.

`Test-ComputerSecureChannel -Verbose`

Validates the computer's trust relationship with its Active Directory domain.

`Test-ComputerSecureChannel -Repair -Credential (Get-Credential) -Verbose`

Repairs a broken computer-domain secure channel.

`Get-NetTCPConnection -State Listen`

Examines TCP ports currently listening on a Windows system.

`pktmon start --capture --comp nics --pkt-size 0`

Captures packets directly from Windows network interfaces for deeper troubleshooting.

Key Learning Points

1. Ping does not prove that an application works

ICMP connectivity only demonstrates basic IP reachability.

A system may successfully return:

```
PingSucceeded : True
```

while:

```
TcpTestSucceeded : False
```

because the required application port is still blocked or incorrectly routed.

2. Active Directory depends heavily on DNS

Domain clients must be able to locate domain controllers through AD-specific DNS records.

Changing CLIENT02 and FILESRV02 to use DC01 as DNS allowed Windows DC Locator to correctly discover `benlab.local`.

3. Active Directory requires multiple network services

Successful LDAP connectivity on TCP 389 did not mean every AD operation would work.

SMB, RPC Endpoint Mapper and dynamically allocated RPC ports were also required during domain operations.

4. Stateful firewall behavior matters

A correct static route does not automatically guarantee that stateful firewall traffic will return through that route.

The pfSense `reply-to` behavior became an important troubleshooting point in this multi-router topology.

5. Packet captures can identify exactly where communication fails

Instead of guessing whether CLIENT02, Windows Firewall, BRANCHFW01, pfSense01 or DC01 was responsible, packet captures allowed traffic to be followed across individual interfaces.

This provided evidence for progressively eliminating potential causes.

6. Troubleshooting should progress from basic connectivity to application validation

A useful troubleshooting sequence from this exercise was:

```
IP Configuration
↓
Ping / Reachability
↓
Routing
↓
ARP / Layer 2
↓
Firewall Rules
↓
Packet Capture
↓
TCP Service Testing
↓
DNS
↓
DC Locator
↓
Secure Channel
↓
Final Application Validation
```

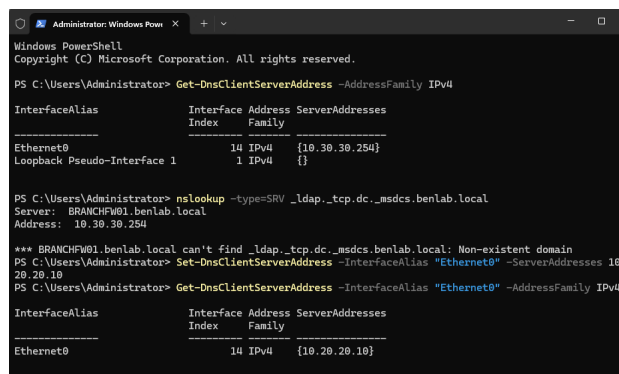
Screenshots

Screenshots — HQ-to-Branch Domain Connectivity Troubleshooting

Screenshot 1 — Branch DNS Misconfiguration Identified

Caption:

Initial DNS and Active
Directory Discovery
Failure



```
Administrator: Windows PowerShell
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PS C:\Users\Administrator> Get-DnsClientServerAddress -AddressFamily IPv4

InterfaceAlias      Interface Address ServerAddresses
-----
Ethernet0           14 IPv4 {10.30.30.254}
Loopback Pseudo-Interface 1  1 IPv4 {}

PS C:\Users\Administrator> nslookup -type=SRV _ldap._tcp.dc._msdcs.benlab.local
Server:  BRANCHFW01.benlab.local
Address: 10.30.30.254

*** BRANCHFW01.benlab.local can't find _ldap._tcp.dc._msdcs.benlab.local: Non-existent domain
PS C:\Users\Administrator> Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses 10.20.20.10
PS C:\Users\Administrator> Get-DnsClientServerAddress -InterfaceAlias "Ethernet0" -AddressFamily IPv4

InterfaceAlias      Interface Address ServerAddresses
-----
Ethernet0           14 IPv4 {10.20.20.10}
```

CLIENT02 was initially configured to use BRANCHFW01 (10.30.30.254) as its DNS server. An SRV lookup for `_ldap._tcp.dc._msdcs.benlab.local` returned Non-existent domain, preventing the workstation from locating an Active Directory domain controller.

Screenshot 2 — DNS Corrected and DC01 Successfully Discovered

Caption:

CLIENT02 DNS Reconfiguration and Domain Controller Discovery

```
Administrator: Windows PowerShell
Loopback Pseudo-Interface 1 1 IPv4 {}

PS C:\Users\Administrator> nslookup -type=SRV _ldap._tcp.dc._msdcs.benlab.local
Server: BRANCHFW01.benlab.local
Address: 10.30.30.254

*** BRANCHFW01.benlab.local can't find _ldap._tcp.dc._msdcs.benlab.local: Non-existent domain
PS C:\Users\Administrator> Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses 10.20.20.10
PS C:\Users\Administrator> Get-DnsClientServerAddress -InterfaceAlias "Ethernet0" -AddressFamily IPv4

InterfaceAlias      Interface Index Address ServerAddresses
-----
Ethernet0           14 IPv4 {10.20.20.10}

PS C:\Users\Administrator> nltest /dsgetdc:benlab.local
DC: \DC01.benlab.local
Address: \\10.20.20.10
Dom Guid: 8c89b666-bb8a-defd-acf2-c99811e8a034
Dom Name: benlab.local
Forest Name: benlab.local
Dc Site Name: Default-First-Site-Name
Our Site Name: Default-First-Site-Name
Flags: PDC GC DS LDAP HDC TIMESERV GTIMESERV WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE
FULL_SECRET WS_DS_8_DS_9_DS_10 KEYLIST
The command completed successfully
PS C:\Users\Administrator>
```

The Ethernet0 DNS configuration was changed to DC01 (10.20.20.10). After the change, `nltest /dsgetdc:benlab.local` successfully located DC01.benlab.local, confirming that CLIENT02 could perform Active Directory domain-controller discovery across the routed Branch-to-HQ connection.

Screenshot 3 — Secure Channel Repair Initially Incomplete

Caption:

Secure Channel Failure Isolated to RPC Dynamic Connectivity

```
Administrator: Windows PowerShell
The command completed successfully
PS C:\Users\Administrator> Test-ComputerSecureChannel -Repair -Credential (Get-Credential) -Verbose

cmdlet Get-Credential at command pipeline position 1
Supply values for the following parameters:
Credential
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "DESKTOP-0499M0A".
True
VERBOSE: The secure channel between the local computer and the domain benlab.local was successfully repaired.
PS C:\Users\Administrator> Test-ComputerSecureChannel -Verbose
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "DESKTOP-0499M0A".
False
VERBOSE: The secure channel between the local computer and the domain benlab.local is broken.
PS C:\Users\Administrator> Test-NetConnection 10.20.20.10 -Port 49152
WARNING: TCP connect to (10.20.20.10 : 49152) failed

ComputerName      : 10.20.20.10
RemoteAddress     : 10.20.20.10
RemotePort        : 49152
InterfaceAlias    : Ethernet0
SourceAddress     : 10.30.30.100
PingSucceeded     : True
PingReplyDetails (RTT) : 2 ms
TcpTestSucceeded  : False
PS C:\Users\Administrator>
```

Although `Test-ComputerSecureChannel -Repair` initially reported a successful repair, subsequent validation returned `False`. A `Test-NetConnection` test to DC01 on TCP port 49152 also failed while ICMP connectivity succeeded, demonstrating that Layer 3 connectivity existed but required Active Directory RPC traffic was still being blocked.

Administrator: Windows PowerShell

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-NetTCPConnection -State Listen |
>> Where-Object { $_.LocalPort -ge 49152 } |
>> Sort-Object LocalPort |
>> Select-Object -First 10 LocalAddress,LocalPort,State

LocalAddress LocalPort State
-----
0.0.0.0      49664 Listen
::          49664 Listen
::          49665 Listen
0.0.0.0      49665 Listen
0.0.0.0      49666 Listen
::          49666 Listen
0.0.0.0      49667 Listen
::          49667 Listen
::          49668 Listen
0.0.0.0      49668 Listen

PS C:\Users\Administrator> _
```

Screenshot 4 — RPC Dynamic Port Range Verified on DC01

Caption:

DC01 RPC Dynamic Port Validation

Get-NetTCPConnection was used on DC01 to confirm that Windows services were actively listening on dynamic RPC ports beginning in the 49152–65535 range. This established that the failed branch connection was not simply caused by the tested port being inactive on the domain controller.

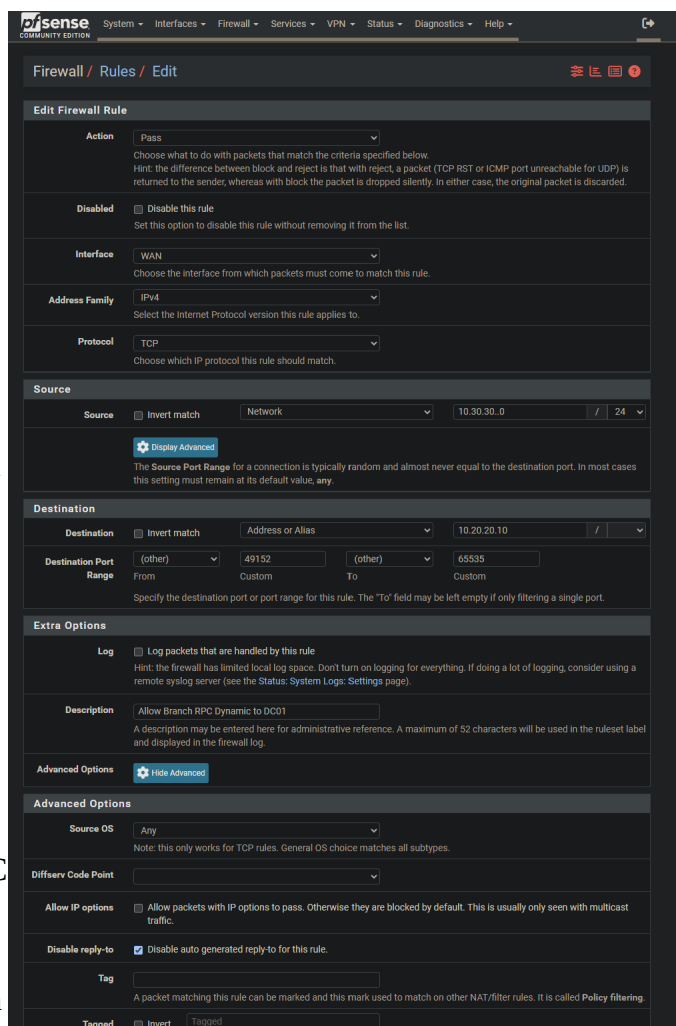
Screenshot 5 — pfSense Inter-Site RPC Rule with Disable reply-to

Caption:

pfSense Branch-to-HQ RPC Dynamic Firewall Rule

A controlled WAN
firewall rule was
configured to permit
TCP traffic from the
Branch network
(10.30.30.0/24) to
DC01 (10.20.20.10)
across the Windows RPC
dynamic port range
49152–65535. The
rule was configured with
Disable reply-to to

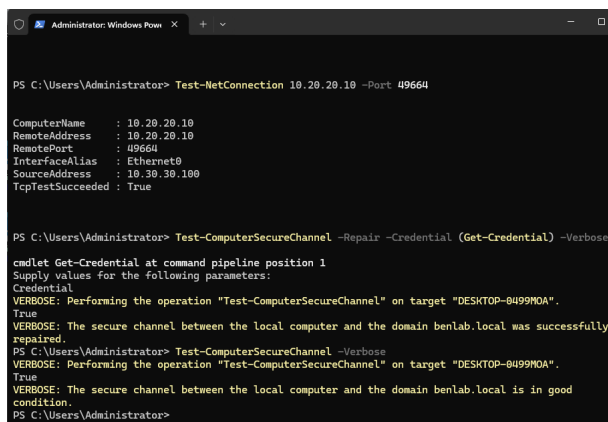
prevent pfSense from forcing return traffic toward the WAN default gateway rather than following the explicit inter-site route.



Screenshot 6 — CLIENT02 RPC Connectivity and Secure Channel Fully Validated

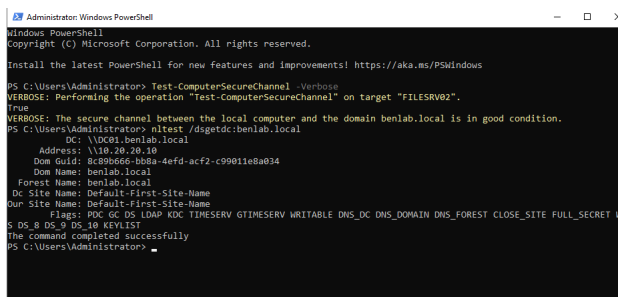
Caption:

CLIENT02 Final Active Directory Connectivity Validation



After the firewall correction, `Test-NetConnection 10.20.20.10 -Port 49664` returned `TcpTestSucceeded : True`. The workstation's domain trust was then repaired and independently retested with `Test-ComputerSecureChannel -Verbose`, which returned `True` and confirmed that the secure channel with `benlab.local` was in good condition.

Screenshot 7 — FILESRV02 Final Branch Server Validation



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Test-ComputerSecureChannel -Verbose
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "FILESRV02".
True
VERBOSE: The secure channel between the local computer and the domain benlab.local is in good condition.
PS C:\Users\Administrator> nltest /dsgetdc:benlab.local
DC: \DC01.benlab.local
Address: \\10.20.20.10
DomainId: 8c9b0d0e-b0ba-4afd-acf2-c99011e8a034
Domain Name: benlab.local
Forest Name: benlab.local
DC Site Name: Default-First-Site-Name
Our Site Name: Default-First-Site-Name
Flags: RODC GC DS LDAP KDC TIMESERV GTIMESERV WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE FULL_SECRET W
S DS_8 DS_9 DS_10 KEV1157
The command completed successfully
PS C:\Users\Administrator>
```

Caption:

FILESRV02 Domain Integration and Secure Channel Validation

Following domain integration of the Branch file server, `Test-ComputerSecureChannel -Verbose` returned `True`. `nltest /dsgetdc:benlab.local` also successfully discovered `DC01.benlab.local` at `10.20.20.10`, confirming that `FILESRV02` could authenticate with the `BENLAB` domain and locate Active Directory services across the routed Branch-to-HQ infrastructure.

Milestone Result

Status: COMPLETED / VALIDATED

The `BENLAB` environment now demonstrates a simulated multi-site enterprise network in which Branch 01 systems communicate across routed pfSense infrastructure with Headquarters and consume centralized Active Directory and DNS services hosted on `DC01`.

Both a Windows workstation (**CLIENT02**) and Windows Server (**FILESRV02**) were successfully validated against the Headquarters `benlab.local` domain.